

- **COMP 2522**
- **Fall 2025**
- **Object Oriented Programming 1**

Quiz 3/ Version 1

1. What is the rule when overriding hashCode() in Java? (1 mark)
2. Create a class Phone with a yearMade variable. Make a getYearMade() method that cannot be overridden. Extend the class with an iPhone class that cannot be extended. iPhones have a String color variable. Phones made in the same year are equal. But iPhones with the same color are equal. (11 marks)

Quiz 3/ Version 2

1. Create a Java interface called Quiz that represents the structure for a basic quiz. The interface must include the following:
 An abstract method called askQuestion() that returns a String representing the quiz question.
 An abstract method called checkAnswer() that takes a String parameter representing a user's answer and returns a boolean indicating whether the answer is correct.
 A default method called provideFeedback() that takes a String parameter (the user's answer) and returns a String indicating whether the answer is correct or incorrect. Use the checkAnswer() method inside the default method to determine the feedback.
 Your interface should define the basic structure for any type of quiz, and classes that implement the interface will define specific quiz behavior.
 Also create a Comp2522Test class which implements Quiz.
 (8 marks)

Quiz 3/ Version 3

1. What is the rule when overriding hashCode() in Java? (1 mark)
2. Create a class Phone with a yearMade variable. Make a getYearMade() method that cannot be overridden. Extend the class with an iPhone class that cannot be extended. iPhones have a String colour variable. Phones made in the same year are equal, but iPhones with the same colour are equal. (11 marks)

3. Explain everything that is wrong with the following code. Circle any errors and explain what is wrong and how to fix it. Use best practices (but ignore JavaDoc here). (8 marks)

```
abstract final class A extends Object
{
    Public private final int x;
    A(final int x)
    {
        This.x = x;
    }
    final void y()
    {
        System.out.println("this is x: " + x);
    }
    int getX()
    {
        return x;
    }
}
```

```
Class B extends A
{
    B(final String z)
    Private final String z;
    {
        super(z.length());
        this.z = z;
    }
    @Override
    Void y()
    {
        System.out.println("hi");
    }
    private boolean equals(final Object o)
    {
        Return this.x == x;
    }
}
```